David Matos

AI Engineer | Software / Backend Engineer

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Summary

AI Engineer and Computer Science graduate building **LLM-powered agents for data-pipeline observability**, with internship experience in **Python backend development, data processing, reusable software, CI/CD, and cloud-native technologies**. Former **professional handball player** with **10 years of competitive team experience**.

Education

Faculty of Sciences, University of Porto — Porto, Portugal
B.Sc. in Computer Science — 2022 – 2026

- Relevant coursework: Software Engineering, Distributed Systems, Algorithms, Databases, Large-Scale Data Science, Machine Learning, and Problem Solving.

Experience

AI Engineer — October 2026 – Present
DataCentric — *Portugal*

- Reduced data engineers' **debugging time** on failed pipeline runs by co-developing an **LLM-powered agent** that monitors **Databricks** job runs and automatically reports **what failed, how it failed, and how to fix it**.
- Enabled **root-cause tracing** of job-run errors back to their origin by building a **Neo4j knowledge graph** of pipelines, jobs, and dependencies, with the agent's logic implemented in **Python, SQL, and YAML**.
- Delivered the solution to an external client as part of a **team of 3 engineers**, contributing across design, implementation, and integration with the client's data platform.

Curricular Intern – Software Engineering — October 2025 – January 2026
Celfinet, a Cyient Company — *Lisbon, Portugal*

- Accelerated data extraction for ad-hoc studies, with the library adopted by **two internal teams** and still in use after the internship, by designing a reusable **Python library** with **hundreds of functions** for direct access to frequently used datasets.
- Reduced the complexity of recurring data-access workflows by building reusable abstractions over internal databases with **SQLAlchemy**, validated by comprehensive **unit tests**.
- Enabled developers to install the library directly by automating its testing, packaging, and internal distribution through a **CI/CD pipeline** in **Azure DevOps**.

Summer Intern – Software Engineering — Summer 2025
Celfinet, a Cyient Company — *Lisbon, Portugal*

- Improved maintainability and component reusability of a **FastAPI backend** for base-station energy consumption studies by refactoring and restructuring its code organization.
- Increased code reuse across data-intensive workflows, with packages adopted by an internal team, by developing reusable **Python components across dozens of internal packages** for processing **very large datasets**.
- Improved the quality of customer-submitted data by building a **CSV validation and error-reporting system** that detects **missing values, invalid data, incorrect units, and duplicates**, and automatically corrects them where reliable rules apply.
- Ensured reliable backend and data-processing behavior by developing and maintaining **automated tests**.

Vice President — 2023 – 2024
NUCC – Student Nucleus of Computer Science — *Faculty of Sciences, University of Porto*

- Organized technical and career-oriented events connecting Computer Science students with industry professionals and external organizations.
- Coordinated communication and execution across **students, faculty members, industry professionals, and external organizations**.
- Managed multiple initiatives within the Computer Science student community, developing practical experience in coordination, communication, and leadership.

Projects

Personal Cloud Platform for Large-Scale Data Science | *Python, Docker, Kubernetes, Linux* 2026

- Designed and implemented a **cloud-based platform for large-scale data science workloads**, using containerized services and cloud-native technologies.
- Used **Docker and Kubernetes** to package, deploy, and orchestrate application components, exploring container orchestration and scalable system deployment.
- Designed the system with a focus on **modularity, scalability, and automated deployment** within a Linux-based environment.

K-Nearest Neighbors (KNN) Model Optimization | *Python, Machine Learning, scikit-learn* 2026

- Implemented and evaluated the **K-Nearest Neighbors (KNN)** algorithm in Python, analyzing model performance across different datasets and configurations.
- Applied **K-Fold cross-validation** to evaluate model generalization and compare different KNN configurations using consistent evaluation procedures.
- Investigated the impact of **feature scaling, PCA dimensionality reduction**, and **Manhattan distance** on model performance and classification results.
- Compared different preprocessing and model configurations, analyzing their impact on **accuracy, generalization, and computational performance**.

Technical Skills

Programming Languages: Python, Java, C, C++, SQL, JavaScript, YAML
AI & LLMs: LLM-based Agents, Knowledge Graphs (Neo4j), Root-Cause Analysis, Prompt Engineering
Backend: FastAPI, Flask, Django, REST APIs, SQLAlchemy
Data Platforms: Databricks, Neo4j, Data Pipelines
Cloud & DevOps: Docker, Kubernetes, CI/CD, Linux, Bash, Azure, GCP
Software Engineering: Object-Oriented Programming, Data Structures & Algorithms, Automated Testing, Git, Version Control
Data & Machine Learning: Data Processing, Data Analysis, Feature Engineering, scikit-learn, Cross-Validation, PCA, Model Evaluation
Tools: Postman
Languages: Portuguese (Native), English (Professional)

Additional Experience

Professional Handball Player 2012 – 2022
Competitive Team Sport *Portugal*

- Competed at a high level for 10 years, developing strong experience in teamwork, discipline, communication, and performance under pressure.